Yunpeng Wang

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Education

University of Pennsylvania Aug 2022–May 2025
M.S.E., Electrical Engineering; M.S.E., Computer and Information Science | GPA: 3.70

Central South University Sep 2018–Jun 2022
B.Eng., Internet of Things Engineering | GPA: 3.77

Experience

WeRide
Software Engineer Jul 2025–Present
- Maintain and extend planning-simulation and scheduling systems used for continuous autonomous-driving evaluation.
- Develop incident-analysis workflows using vision-language models and decision-tree classifiers.

Software Engineer Intern Jun 2024–Dec 2024
- Built a microservice simulation launcher for PnC CI; improved preprocessing throughput 40% and cut memory use 42%.

Alibaba Cloud *Software Engineering Intern* Jun 2023–Sep 2023
- Built an unsupervised-learning anomaly-detection service and operations dashboard for a cloud storage cluster.

Research

xLab, GRASP Laboratory, University of Pennsylvania *Research Assistant* Jan 2025–May 2025
- Developed and evaluated an F1TENTH sim-to-real RL pipeline: Pure Pursuit imitation warm-up, then SAC/PPO with LSTM policies to capture unobserved state, domain randomization, and two-car overtaking in simulation.
- Deployed to the physical car: the domain-randomized policy lapped in 26 s. [code] [writeup]

Projects

full writeups at happywalkers.github.io

Robotics & Deep Learning

Drone path planning and control: 26-connected A* search, map inflation, path simplification, and geometric PID control; implemented and timed fifth-order minimum-jerk trajectory segments for the visual-inertial extension.

Robot-arm block stacking: built block detection and inverse kinematics; team 1st in class.

Kalman filters: augmented-state EKF for parameter estimation; quaternion UKF for IMU orientation.

LiDAR particle-filter SLAM: 1,000 particles, scan-to-map scoring, stratified resampling, 0.05 m occupancy grid.

NeRF: from COLMAP poses: ray sampling, sine coordinate features, a three-layer MLP, and differentiable compositing.

PPO walker: PPO on DeepMind Control with GAE and KL early stopping; trained a walking policy in simulation.

Binary weight quantization: $\{-1, +1\}$ vs. $\{0, 1\}$ weights, deterministic vs. stochastic; deterministic ± 1 scored best.

Positional embeddings in Transformers: derived additive encoding as a constrained form of concatenation.

Thesis quality prediction: predicted essay quality from users' writing behavior with BERT and an LSTM.

Systems

Distributed web search engine: distributed key-value store, RDD-style distributed processing, crawler, inverted index, TF-IDF/PageRank ranking, and a query interface; deployed on 16 EC2 instances, crawling and ranking 100k+ pages.

Raft consensus: leader election, log replication, persistence, snapshot compaction, and `InstallSnapshot` recovery.

Chord DHT: joins, stabilization, finger tables, successor lookup, failure detection, and a callback registry for async RPCs.

BusTub relational database: ARC buffer pool and disk scheduler, B+tree with optimistic latch crabbing, batched executors with external merge sort, and MVCC with snapshot isolation, garbage collection, and serializable validation.

JOS operating system: virtual memory, user environments and traps, copy-on-write fork, preemptive round-robin scheduling, and IPC; a file server, block cache, and E1000 NIC driver using MMIO and DMA descriptor rings.

Also: Oat compiler (LLVM frontend, x86-64 backend, register allocation) · FinanceTracker, a self-hosted bookkeeping app on a local LLM (FastAPI, PostgreSQL, Docker) · course-review web app (Django, then Spring Boot and Angular) · team smart-home system (Raspberry Pi, STM32) · water-meter reading research at Central South University (MSER, NMS, SURF)

Honors, Credentials, and Skills

Awards: 1st of 150 teams, Hunan Collegiate Programming Contest (2021) · Meritorious Winner, MCM/ICM (2021)
IP (China): patent application 202111412755.9 (wireless weighing for vending machines)
Languages: Python, C++, C, Go, Java, SQL · **Robotics:** ROS/ROS 2, DeepMind Control, MuJoCo
ML: PyTorch, Transformers, Stable-Baselines3 · **Infra:** Docker, Kubernetes, Bazel, AWS EC2, PostgreSQL, Redis